BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: PR X (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR X (01) 21

Note: Four circles are given against each question. Fill only the circle of the correct answer. Use a blue or black marker/pen. Filling more than one circle will be treated as a wrong answer.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The equilibrium constant (Kc) is:

(A) Sum of two reactions
(B) Difference of two rate constants
(C) Ratio of two rate constants
(D) Product of two rate constants

Q2. Lactic acid is mostly found in:

(A) Lemons
(B) Insects
(C) Spinach
(D) Sour milk

Q3. Which one of the given is a Lewis base?

(A) HCl
(B) $AlCl_3$
(C) BF_3
(D) F^-

Q4. Which of the given compounds responds to CnH2n+2 formula?

(A) C_2H_2
(B) C_2H_4
(C) C_2H_6
(D) C_2H_3

Q5. Oxygen is attached on both sides to carbon atom in:

(A) Alcohol
(B) Ketone
(C) Aldehyde
(D) Ether

Q6. Baeyer's test is commonly used for the detection of:

(A) Cycloalkanes
(B) Aromatic compounds
(C) Alkanes
(D) Alkenes

Q7. On hydrolysis lactose gives glucose and:

(A) Sucrose
(B) Galactose
(C) Maltose
(D) Fructose

Q8. Stearic acid ($C_{17}H_{35}COOH$) is found in:

(A) Butter
(B) Animal fat
(C) Vinegar
(D) Vitamins

Q9. Which pair of gases is found in atmosphere with 99% of amount?

(A) Carbondioxide and Neon
(B) Helium and Hydrogen
(C) Argon and Hydrogen
(D) Nitrogen and oxygen

Q10. The coldest region in atmosphere is:

(A) Troposphere
(B) Stratosphere
(C) Mesosphere
(D) Thermosphere

Q11. Chlorine acts as bleaching agent in the presence of:

(A) HCl
(B) H_2O
(C) NaOH
(D) HNO_3

Q12. The impurity present in mineral is called:

(A) Gangue
(B) Mineral
(C) Ore
(D) Metal

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. In the equilibrium mixture, the concentration of hydrogen and iodine is 0.04 moles per dm^3 each while that of hydrogen iodide is 0.08 moles per dem^3. Find Kc of the given reaction: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$.

2. Discuss the concept of Lewis acids and bases with examples.

3. What are alkanes? Why alkanes are also called "Paraffins"?

4. Define function group. Write at least three function groups containing oxygen.

5. Write balance equations for the preparation of ethene from ethyl alcohol and ethyl chloride.

6. Differentiate between glucose and fructose.

7. Sketch and briefly discuss the composition of atmosphere by mentioning the percentage of each component.

8. Explain hydrogen bonding in water with the help of structure of water molecules.

9. How blister copper is purified?

10. Write any four physical properties of water.

11. Compare the different macroscopic characteristics of forward and reversible reactions.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] Differentiate between different organic compounds on the basis of alcoholic and ether functional groups.

(b) [3] Discuss any three effects of ozone layer depletion.

QIV.

(a) [4] Define hard and soft water and discuss the causes of hardness of water.

(b) [3] Complete the reactions: i) Sucrose --H_2O--> Glucose + ? ii) Lactose --H_2O--> Glucose + ? iii) Maltose --H_2O--> Glucose + ?

QV.

(a) [4] Explain the graphical presentation of dynamic equilibrium with its proper example.

(b) [3] Calculate the pOH and pH of 0.01 M solution of potassium hydroxide (KOH).

QVI.

(a) [4] What is petroleum. Name any three fractions showing the number of carbon atoms per molecules.

(b) [3] Write a note on stratosphere.